# Run report: item4-run-20261018T2049Z

Descriptive record of one recorded run of examples/genus4_res_census.rs (Parts T, H4, K) under `research/weil_descent_angles_20261018/PROTOCOL.md` (Addendum A and Amendments B–G, each committed before the run it governs). It lists what ran, on what host, and what the tool recorded, file by file. Interpretation is in the study's `REPORT.md`. Costs are counted operations ($F_p$ multiplications and inversions, group operations, native operations), never wall-clock; the only times are the tool's own seconds. Isolation: none earned (a shared cloud container, 4 threads); this affects only the seconds, since every recorded value is an exact count or a counted operation total.

## env.txt

```
run: item 4, 20261018T2049Z (Parts T, H4, K)
protocol commit: bdcf8f8ad (item 4 frozen); tool commit: 252b52998
rustc 1.97.0 (2d8144b78 2026-07-07)
host: Linux 6.18.44-fc-v139 x86_64, 4 cores, Intel(R) Xeon(R) Processor @ 2.10GHz, 15 GB
command: rustc --edition 2021 -O examples/genus4_res_census.rs -o g4 && G4_THREADS=4 ./g4 research/weil_descent_angles_20261018/results/item4-run-20261018T2049Z T H4 K
tool sha256: 8c6597ea8d5eb73604f8f2d3d724df4275deefce75db8944c6dc2c2b6edcc27b
threads: 4
smoke run before this run (not evidence): G4_LIMIT=200000 (4 s), Parts T and H4 as here; the K slice showed the expected orbit-outside-slice K5 failures only.
```

## tool_log.txt

```
partT: n with an odd-t solution and n >= 3: [] (P5 predicts none)
partT: n = 2 odd-t cycle types: [6] [3, 3]
partT: n = 4 even-t solutions all in powers of 2: true
field F_3^12 ready 0.2s
H4: squarefree degree-10 binary forms = 104976; mass = 104976/48 = 2187 (check K1 expects 2187: ok)
H4: Res-shaped models = 1100; t histogram {-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
H4: check K2 (t even and cycle types in powers of 2) violations = 0
classify split: 36 classes, K5 violations 0
part K split: forms 43046720 smooth 21830400 mass 9475.0000 res_forms 81216 classes 36 (416.4s)
classify nonsplit: 39 classes, K5 violations 0
part K nonsplit: forms 43046720 smooth 27299520 mass 9479.0000 res_forms 110880 classes 39 (714.3s)
classify cone: 83 classes, K5 violations 0
part K cone: forms 14348906 smooth 8503056 mass 6561.0000 res_forms 101520 classes 83 (267.0s)
K: non-hyperelliptic mass total = 25515.000000 (check K3 literature value q^9+q^8-q^6 = 25515: match)
K: Res-shaped classes = 158; t histogram (class counts) {-18: 3, -17: 2, -16: 3, -15: 1, -14: 8, -13: 6, -12: 2, -11: 7, -10: 3, -9: 1, -8: 14, -7: 3, -6: 11, -5: 9, -4: 5, -3: 15, -2: 11, -1:
2, 0: 4, 1: 4, 2: 17, 3: 2, 4: 9, 5: 2, 7: 6, 9: 3, 10: 4, 11: 1}
K: check K4 violations = 0; check K5 violations = 0
K: P6 prime-order classes (82 - t prime, elliptic trace exists) = 9 : ["split:t=11:#E=71:aut=1", "nonsplit:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=9:#E=73:aut=3",
"cone:t=9:#E=73:aut=1", "cone:t=9:#E=73:aut=1", "cone:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=-7:#E=89:aut=1"]
total 1398.1s
```

## checks.txt

```
partT: n with an odd-t solution and n >= 3: [] (P5 predicts none)
partT: n = 2 odd-t cycle types: [6] [3, 3]
partT: n = 4 even-t solutions all in powers of 2: true
H4: squarefree degree-10 binary forms = 104976; mass = 104976/48 = 2187 (check K1 expects 2187: ok)
H4: Res-shaped models = 1100; t histogram {-18: 18, -10: 96, -8: 144, -6: 96, -2: 192, 0: 204, 2: 180, 4: 48, 8: 48, 10: 48, 16: 24, 18: 2}
H4: check K2 (t even and cycle types in powers of 2) violations = 0
K: non-hyperelliptic mass total = 25515.000000 (check K3 literature value q^9+q^8-q^6 = 25515: match)
K: Res-shaped classes = 158; t histogram (class counts) {-18: 3, -17: 2, -16: 3, -15: 1, -14: 8, -13: 6, -12: 2, -11: 7, -10: 3, -9: 1, -8: 14, -7: 3, -6: 11, -5: 9, -4: 5, -3: 15, -2: 11, -1:
2, 0: 4, 1: 4, 2: 17, 3: 2, 4: 9, 5: 2, 7: 6, 9: 3, 10: 4, 11: 1}
K: check K4 violations = 0; check K5 violations = 0
K: P6 prime-order classes (82 - t prime, elliptic trace exists) = 9 : ["split:t=11:#E=71:aut=1", "nonsplit:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=9:#E=73:aut=3",
"cone:t=9:#E=73:aut=1", "cone:t=9:#E=73:aut=1", "cone:t=-1:#E=83:aut=1", "cone:t=-7:#E=89:aut=1", "cone:t=-7:#E=89:aut=1"]
```

## h4_res_shaped.csv (first 12 of 1100 rows)

| code | a0..a10 | N4 | t | t_even | cycle_type | cycles_powers_of_2 |
|---|---|---|---|---|---|---|
| 19686 | 0 1 0 0 0 0 0 0 0 1 0 | 154 | −18 | true | 1+1+4+4 | true |
| 19687 | 1 1 0 0 0 0 0 0 0 1 0 | 154 | −18 | true | 1+1+4+4 | true |
| 19688 | 2 1 0 0 0 0 0 0 0 1 0 | 154 | −18 | true | 1+1+4+4 | true |
| 19689 | 0 2 0 0 0 0 0 0 0 1 0 | 10 | 18 | true | 1+1+1+1+2+2+2 | true |

| | | | | | | |
|---|---|---|---|---|---|---|
| 19716 | 0 2 0 1 0 0 0 0 0 1 0 | 82 | 0 | true | 1+1+8 | true |
| 19717 | 1 2 0 1 0 0 0 0 0 1 0 | 82 | 0 | true | 1+1+8 | true |
| 19718 | 2 2 0 1 0 0 0 0 0 1 0 | 82 | 0 | true | 1+1+8 | true |
| 19743 | 0 2 0 2 0 0 0 0 0 1 0 | 82 | 0 | true | 1+1+8 | true |
| 19744 | 1 2 0 2 0 0 0 0 0 1 0 | 82 | 0 | true | 1+1+8 | true |
| 19745 | 2 2 0 2 0 0 0 0 0 1 0 | 82 | 0 | true | 1+1+8 | true |
| 19932 | 0 2 0 0 0 1 0 0 0 1 0 | 18 | 16 | true | 1+1+8 | true |
| 19960 | 1 2 0 1 0 1 0 0 0 1 0 | 90 | −2 | true | 1+1+8 | true |

## k_classes.csv (first 12 of 158 rows)

| type | rep_code | orbit_forms | aut | N4 | t | numE | numE_prime | elliptic_trace_exists | form |
|---|---|---|---|---|---|---|---|---|---|
| split | 708734 | 2304 | 1 | 94 | −3 | 85 | false | false | 2*x^0y^0 + 1*x^0y^2 + 2*x^0y^3 + 1*x^1y^0 + 1*x^2y^3 + 1*x^3y^0 |
| split | 708761 | 2304 | 1 | 106 | −6 | 88 | false | false | 2*x^0y^0 + 1*x^0y^2 + 2*x^1y^0 + 1*x^2y^3 + 1*x^3y^0 |
| split | 708989 | 2304 | 1 | 106 | −6 | 88 | false | false | 2*x^0y^0 + 1*x^0y^1 + 2*x^0y^2 + 2*x^0y^3 + 1*x^1y^0 + 1*x^1y^1 + 1*x^2y^3 + 1*x^3y^0 |
| split | 728311 | 2304 | 1 | 114 | −8 | 90 | false | true | 1*x^0y^0 + 1*x^0y^1 + 1*x^0y^2 + 1*x^0y^3 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0 |
| split | 729470 | 2304 | 1 | 90 | −2 | 84 | false | true | 2*x^0y^0 + 1*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^1 + 1*x^1y^2 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0 |
| split | 729949 | 2304 | 1 | 102 | −5 | 87 | false | true | 1*x^0y^0 + 1*x^0y^1 + 2*x^0y^3 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^1 + 1*x^2y^3 + 1*x^3y^0 |
| split | 755479 | 2304 | 1 | 54 | 7 | 75 | false | true | 1*x^0y^0 + 2*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^2 + 1*x^2y^0 + 2*x^2y^1 + 1*x^2y^3 + 1*x^3y^0 |
| split | 768113 | 2304 | 1 | 78 | 1 | 81 | false | true | 2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 1*x^1y^1 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0 |
| split | 768506 | 2304 | 1 | 94 | −3 | 85 | false | false | 2*x^0y^0 + 1*x^0y^1 + 2*x^0y^3 + 1*x^1y^0 + 1*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0 |
| split | 769274 | 2304 | 1 | 66 | 4 | 78 | false | true | 2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0 |
| split | 769336 | 2304 | 1 | 94 | −3 | 85 | false | false | 1*x^0y^0 + 2*x^0y^1 + 2*x^0y^2 + 2*x^0y^3 + 2*x^1y^0 + 2*x^1y^2 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0 |
| split | 774323 | 2304 | 1 | 38 | 11 | 71 | true | true | 2*x^0y^0 + 2*x^0y^1 + 1*x^0y^2 + 1*x^0y^3 + 1*x^1y^0 + 1*x^2y^0 + 1*x^2y^2 + 1*x^2y^3 + 1*x^3y^0 |

## k_summary.csv (first 3 of 3 rows)

| type | forms | smooth | mass_den | mass | smooth_with_N1_4 | res_shaped_forms | res_shaped_classes | k5_violations | seconds |
|---|---|---|---|---|---|---|---|---|---|
| split | 43046720 | 21830400 | 2304 | 9475.000000 | 4100256 | 81216 | 36 | 0 | 416.4 |
| nonsplit | 43046720 | 27299520 | 2880 | 9479.000000 | 5611680 | 110880 | 39 | 0 | 714.3 |
| cone | 14348906 | 8503056 | 1296 | 6561.000000 | 1925208 | 101520 | 83 | 0 | 267.0 |

## part_t.csv (first 12 of 30 rows)

| n | t_parity | solutions | cycle_types |
|---|---|---|---|
| 2 | odd | 2 | 6 3+3 |
| 2 | even | 6 | 4+2 4+1+1 2+2+2 2+2+1+1 2+1+1+1+1 1+1+1+1+1+1 |
| 3 | odd | 0 | |
| 3 | even | 4 | 6+2 6+1+1 3+3+2 3+3+1+1 |
| 4 | odd | 0 | |
| 4 | even | 14 | 8+2 8+1+1 4+4+2 4+4+1+1 4+2+2+2 4+2+2+1+1 4+2+1+1+1+1 4+1+1+1+1+1+1 2+2+2+2+2 2+2+2+2+1+1 2+2+2+1+1+1+1 2+2+1+1+1+1+1+1 2+1+1+1+1+1+1+1+1 1+1+1+1+1+1+1+1+1+1 |
| 5 | odd | 0 | |

| | | | |
|---|---|---|---|
| 5 | even | 4 | 10+2 10+1+1 5+5+2 5+5+1+1 |
| 6 | odd | 0 | |
| 6 | even | 8 | 12+2 12+1+1 6+6+2 6+6+1+1 6+3+3+2 6+3+3+1+1 3+3+3+3+2 3+3+3+3+1+1 |
| 7 | odd | 0 | |
| 7 | even | 4 | 14+2 14+1+1 7+7+2 7+7+1+1 |

**Graph**

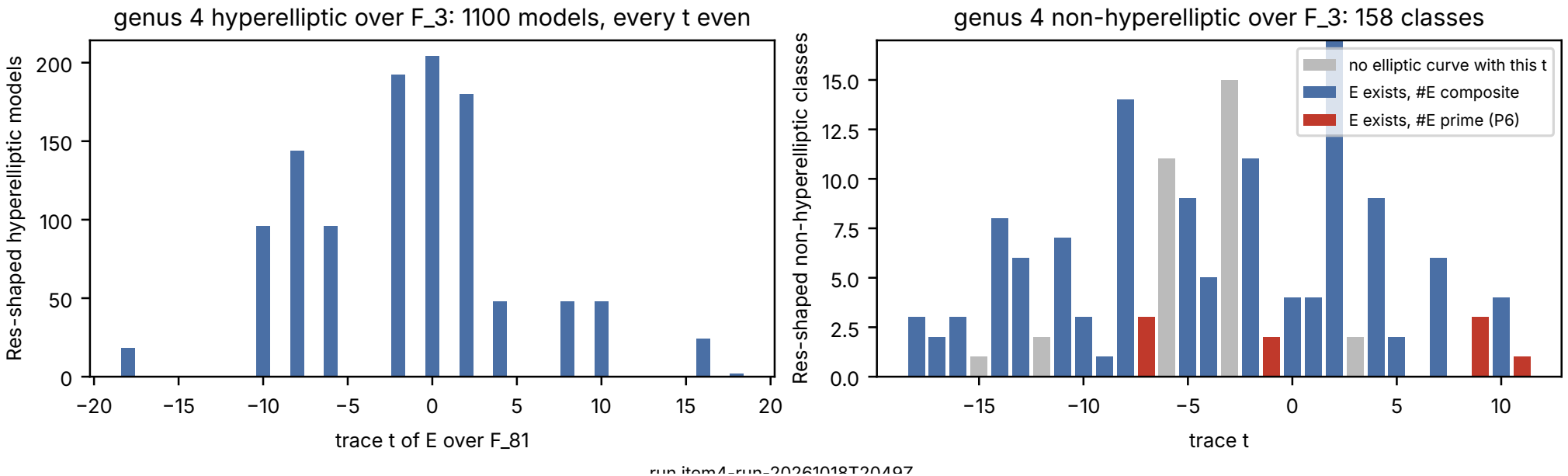



Figure 1: Res-shaped genus-4 curves over F_3 by trace: hyperelliptic models (left), non-hyperelliptic classes (right); exact census.